

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 14)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

Table 1 shows three specialist solid wastes.

1

Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

2

Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

3

Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

01 . 4

Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

10

02 . 1

Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

0 2 . 2

Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0 2 . 3

Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

1

2

3

10

0 3

Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO ₂ emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

0 3 . 1

Explain how nuclear fission releases energy in a reactor.

[3 marks]

03 . 2

Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

03 . 3

Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

03 . 4

Explain why the CO₂ emissions of nuclear power are not zero.

[1 mark]

03 . 5

Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

1

2

10

0 4

Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 4**. 1**

Define the term **half-life**.

[1 mark]**0 4****. 2**

A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

Answer

Bq

0 4**. 3**

Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

04 . 4

Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

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04 . 5

Explain what is meant by an **activation product**.

[1 mark]

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10

05 . 1

Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

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05 . 2

Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

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0 5 . 3

Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

1

2

3

0 5 . 4

Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

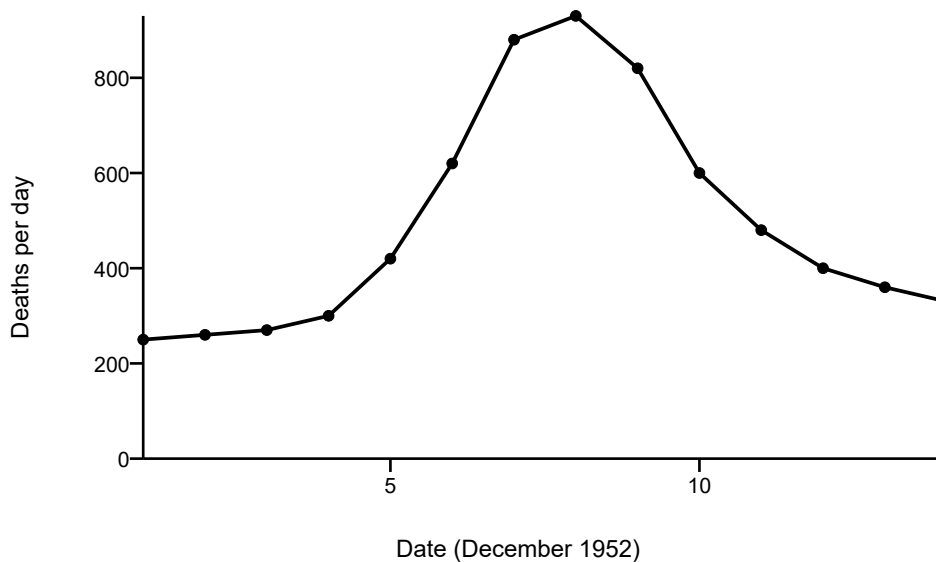
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10

0 6

Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1**Table 1**

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 6**1**

Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 6**2**

Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]**Answer**
 %

06 . 3

Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

06 . 4

Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

06 . 5

State the legislation introduced in the UK in response to this smog.

[1 mark]

10

07 . 1

Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics

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Reduced mass

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Regenerative braking (KERS)

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07 . 2

Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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07 . 3

Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

1

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2

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10

0 8

A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 8

1

Use the formula $D = N(N - 1) / \sum n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 8

2

Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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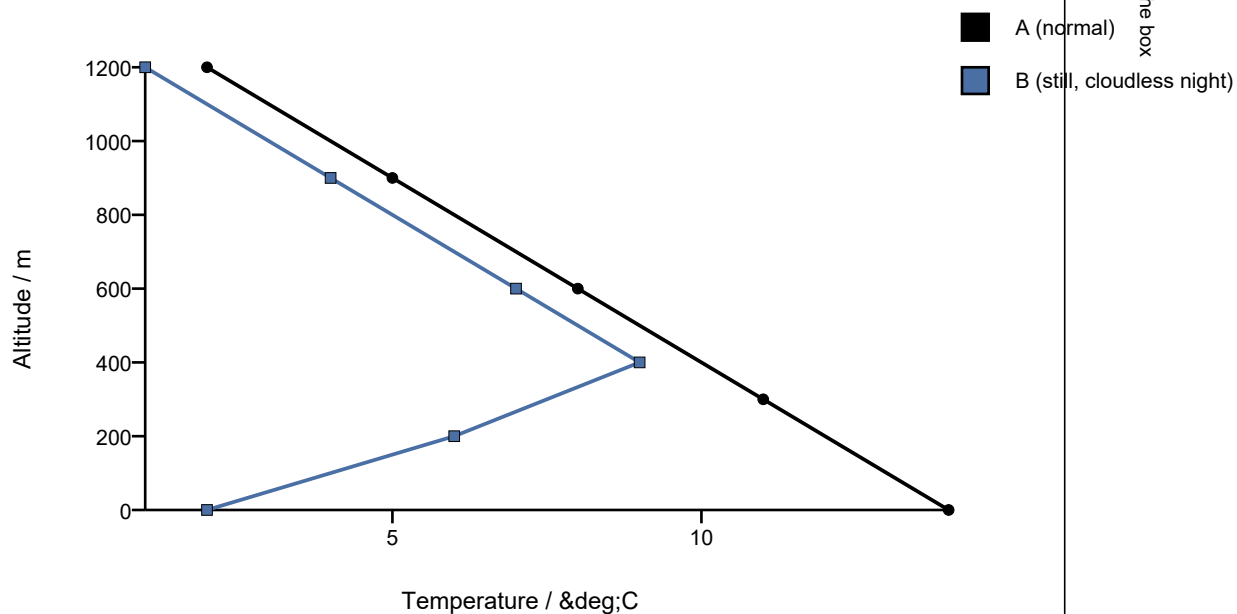
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5

09

Figure 1 shows how air temperature changes with altitude above a city in a valley on two different nights. Line A shows normal conditions. Line B shows the conditions on a still, cloudless night.

Figure 1



09

1 Name the atmospheric condition shown by line B in Figure 1.

[1 mark]

09

2 Use Figure 1 to explain why pollutants released at ground level would disperse on night A but not on night B.

[4 marks]

0	9	.	3
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Explain why the conditions on night B (still, cloudless) allowed an inversion to form.

[3 marks]

0	9	.	4
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Give **two** types of pollution event that are made more severe by temperature inversions.

[2 marks]

10

1	0	.	1
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As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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Extra space

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10 . **2** State what is meant by the **stock** (resource base) of a mineral. **[1 mark]**

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10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2

To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS